Bosung Kim

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Education

University of California, San Diego — Present
PhD in Computer Science and Engineering — San Diego, CA, USA
- Advisor: Prithviraj (Raj) Ammanabrolu

Sogang University — Feb. 2020
M.S. in Computer Science and Engineering — Seoul, South Korea
- Natural Language Processing Lab, advised by Jungyun Seo

Sogang University — Feb. 2016
B.S. in Computer Science and Engineering — Seoul, South Korea
- Magna Cum Laude

Publications

Bridging Frontier Reasoning and Robot Execution: From Autonomous Demonstration Generation to Dense Language Supervision.
Bosung Kim, Alexander Trevithick, Ruiyi Wang, Prithviraj Ammanabrolu
Arxiv, 2026. [paper]

How to Instruct Your Robot: Dense Language Annotations Power Robot Policy Learning.
Bosung Kim*, Ruiyi Wang*, David Acuna, Jaehun Jung, Alexander Trevithick, Brandon Cui, Yejin Choi, Prithviraj Ammanabrolu (*Equal contribution)
NeurIPS, 2026. [paper]

DECADE: Evaluating Evolving Long-Term Memory in Personalized Conversational AI Assistants.
Bosung Kim, Hao Cheng, Yuan Gao, Prithviraj Ammanabrolu, Michel Galley, Rachel Rong, Jianfeng Gao
Under Review, 2026.

Beyond Needle(s) in the Embodied Haystack: Environment, Architecture, and Training Considerations for Long Context Reasoning.
Bosung Kim, Prithviraj Ammanabrolu
CVPR Workshop FMEA, 2025. [paper]

SYMPTOMIFY: Transforming Symptom Annotations with Language Model Knowledge Harvesting.
Bosung Kim, Ndapa Nakashole.
EMNLP Findings, 2023. [paper]

Zero-shot Triplet Extraction by Template Infilling.
Bosung Kim, Hayate Iso, Nikita Bhutani, Estevam Hruschka, Ndapa Nakashole, Tom Mitchell.
IJCNLP-AACL, 2023. [paper]

Chride at SemEval-2023 Task 10: Fine-tuned DebertaV3 on Detection of Online Sexism with Hierarchical Loss.
Letian Peng, Bosung Kim.
ACL Workshop SemEval, 2023. [paper]

Data Augmentation for Rare Symptoms in Vaccine Side-Effect Detection.
Bosung Kim, Ndapa Nakashole.
ACL Workshop BioNLP, 2022. [paper]

Query Reformulation for Descriptive Queries of Jargon Words Using a Knowledge Graph based on a Dictionary.
Bosung Kim, Hyewon Choi, Haeun Yu, Youngjoong Ko.
CIKM, 2021. [paper]

Commonsense Knowledge Augmentation for Low-Resource Languages via Adversarial Learning.
Bosung Kim*, Juae Kim*, Youngjoong Ko, Jungyun Seo. (*Equal contribution)
AAAI, 2021. [paper]

Multi-task learning for Knowledge Graph Completion using Pre-trained Language Models.
Bosung Kim, Taesuk Hong, Youngjoong Ko, Jungyun Seo.
COLING, 2020. [paper]

Publications -Journal

Novel Regularization Method for the Class Imbalance Problem
Bosung Kim, Youngjoong Ko, Jungyun Seo.
Expert Systems With Applications, Vol 188, February 2022.

Work Experience

Research Intern, NVIDIA — Feb. 2026 - June. 2026
NVIDIA Research — Santa Clara, CA, USA
- Mentors: Yejin Choi

Research Intern, Microsoft Research — June. 2025 - Sep. 2025
Deep Learning Group — Redmond, WA, USA
- Project: Agentic Memory Retrieval with Hierarchical Memory System.
- Mentors: Hao Cheng, Michel Galley, Yuan Gao, and Rachel Rong

Applied Scientist Intern, AWS AI Lab — June. 2024 - Sep. 2024
Amazon Q — New York, NY, USA
- Project: Multi-domain Query-driven Knowledge Graph Construction.
- Mentors: Minh Nguyen and Bonan Min

Research Intern, Megagon Labs — June. 2022 - Sep. 2022
Natural Language Processing team — Mountain View, CA, USA
- Project: Zero-shot Relation Extraction using LLMs.
- Mentors: Hayate Iso, Nikita Bhutani, Estevam Hruschka, and Tom Mitchell

Software Engineer, Coupang — Dec. 2015 - Dec. 2017
Query Understanding team — Seoul, South Korea / Mountain View, CA, USA
- Main task: Query processing and query recommendation.

Teaching Assistant	Introduction to Deep Reinforcement Learning, *UC San Diego*	2026 Spring
	AI Agents, *UC San Diego*	2026 Winter
	Accelerated Introduction to Programming, *UC San Diego*	2025 Fall
	Introduction to Deep Reinforcement Learning, *UC San Diego*	2025 Spring
	AI Agents, *UC San Diego*	2025 Winter
	Statistical Natural Language Processing, *UC San Diego*	2024 Fall
	Probabilistic Reasoning and Learning, *UC San Diego*	2023 Fall
	Statistical Natural Language Processing, *UC San Diego*	2023 Spring
	Advanced Natural Language Processing, *Samsung Electronics*	June, Aug 2020
	Advanced Natural Language Processing, *Samsung Electronics*	June, Aug 2019